

Homotopy Pushouts and Homology

Algebraic Topology I | Fall 2026 | October 12 and 14, 2026

Give complete proofs. The preservation of homotopy pushouts by singular chains, the resulting Mayer-Vietoris sequence, and excision may be used after they have been proved in lecture. Coefficients lie in an arbitrary abelian group A , fixed throughout. For a subspace $X' \subseteq X$, relative chains mean the quotient $C_*(X, X'; A) = C_*(X; A) / C_*(X'; A)$.

Exercise 1. Homotopy Pushouts of Chain Complexes

Let K_*, L_*, M_*, N_* be chain complexes of abelian groups, with a square

$$\begin{array}{ccc} K_* & \xrightarrow{f} & L_* \\ g \downarrow & & \downarrow h \\ M_* & \xrightarrow{i} & N_* \end{array}$$

and a degree-one homomorphism $T : K_* \rightarrow N_{*+1}$ satisfying $dT + Td = ig - hf$. Put $Q_* = \text{Cone}((f, g) : K_* \rightarrow L_* \oplus M_*)$, so that

$$Q_n = L_n \oplus M_n \oplus K_{n-1}, \quad d(x, y, z) = (dx + fz, dy + gz, -dz).$$

- Show that $\Phi : Q_* \rightarrow N_*$, defined by $\Phi(x, y, z) = -hx + iy + Tz$, is a chain map. If T' is another such homotopy and $T' - T = dS - Sd$ for a degree-two homomorphism $S : K_* \rightarrow N_{*+2}$, construct a chain homotopy between the two comparison maps.
- Suppose Φ is a quasi-isomorphism, so the square is a homotopy pushout in Ch. Derive the exact sequence

$$\cdots \rightarrow H_n(K) \xrightarrow{(f_*, g_*)} H_n(L) \oplus H_n(M) \xrightarrow{i_* - h_*} H_n(N) \xrightarrow{\delta} H_{n-1}(K) \rightarrow \cdots$$

Here $(i_* - h_*)([x], [y]) = i_*[y] - h_*[x]$. Prove that $\delta[\Phi(x, y, z)] = -[z]$ for each cycle (x, y, z) in Q_n , and verify independence of the representative.

- Suppose the complexes arise by applying $C_*(-; A)$ to a homotopy pushout square of spaces W, X, Y, Z , with maps f, g, h, i in the same positions. Explain which geometric homotopy supplies T and which lecture theorem makes Φ a quasi-isomorphism. Deduce the corresponding long exact sequence in singular homology.

Exercise 2. Spheres, Disks, and Relative Classes

For $n \geq 1$, cover S^n by the complements U and V of its north and south poles. Their intersection deformation retracts onto the equatorial S^{n-1} .

- Use reduced Mayer-Vietoris to compute $\widetilde{H}_k(S^n; A)$ for all $k \geq 0$, starting with the augmentation $H_0(S^0; A) \rightarrow A$. Give the unreduced groups as well, including the case $n = 0$.
- Use the long exact sequence of the pair $S^{n-1} \subseteq D^n$ to compute $H_k(D^n, S^{n-1}; A)$ for every k . Identify the connecting isomorphism to $\widetilde{H}_{n-1}(S^{n-1}; A)$, including $n = 1$.

- (c) For an open cover $X = U \cup V$, use the convention that $C_*(U; A) \oplus C_*(V; A) \rightarrow C_*(X; A)$ sends (x, y) to $y - x$. After subdivision, write a cycle as $z = y - x$ with x supported in U and y in V . Show that $\partial x = \partial y$ is supported in $U \cap V$ and that the Mayer-Vietoris connecting homomorphism sends $[z]$ to $[\partial x]$. Apply this description to the sphere cover to relate the connecting class to an equatorial class. Explain the effect of interchanging U and V .

Exercise 3. Products with the Circle and Tori

Let Y be a nonempty space and cover S^1 by two contractible open arcs whose intersection has two contractible components. Use the resulting cover of $Y \times S^1$.

- (a) Identify the maps from $H_k(Y; A) \oplus H_k(Y; A)$ to the homology groups of the two pieces. With the signs from Exercise 1, show that the first Mayer-Vietoris map is $(a, b) \mapsto (a + b, a + b)$. Determine its kernel and cokernel.
- (b) Deduce a short exact sequence

$$0 \rightarrow H_k(Y; A) \xrightarrow{j_*} H_k(Y \times S^1; A) \rightarrow H_{k-1}(Y; A) \rightarrow 0,$$

where $j : Y \rightarrow Y \times S^1$ includes a chosen slice and $H_{-1}(Y; A) = 0$. Use the projection $p : Y \times S^1 \rightarrow Y$ to prove that the sequence splits. Specify the identifications of the kernel and cokernel used to obtain this sequence.

- (c) For $m \geq 0$, let $T^m = (S^1)^m$, with T^0 a point. Compute $H_k(T^m; A)$ by induction. State the groups of T^2 and T^3 explicitly. For integral coefficients, compute the Euler characteristic of T^m when $m > 0$.

Exercise 4. Complex Projective Spaces

Write $\mathbb{CP}^n$ for the space of complex lines in $\mathbb{C}^{n+1}$, with $\mathbb{CP}^{n-1}$ the subspace on which the last homogeneous coordinate vanishes. For $n \geq 1$, regard D^{2n} as the unit disk in $\mathbb{C}^n$ and define

$$f : D^{2n} \rightarrow \mathbb{CP}^n, \quad f(z_0, \dots, z_{n-1}) = [z_0 : \dots : z_{n-1} : \sqrt{1 - \|z\|^2}].$$

- (a) Show that the restriction to S^{2n-1} has image in $\mathbb{CP}^{n-1}$ and that f identifies the interior of the disk homeomorphically with the complement of $\mathbb{CP}^{n-1}$. Prove that the resulting square attaching D^{2n} along S^{2n-1} is a pushout. Explain why it is a homotopy pushout, using that the sphere inclusion is a Hurewicz cofibration.
- (b) Use excision for this cell attachment to identify $H_k(\mathbb{CP}^n, \mathbb{CP}^{n-1}; A)$ with $H_k(D^{2n}, S^{2n-1}; A)$. Starting with $\mathbb{CP}^0$ a point, use the long exact sequence of the pair to compute $H_k(\mathbb{CP}^n; A)$ for all k . State which maps induced by $\mathbb{CP}^{n-1} \hookrightarrow \mathbb{CP}^n$ are isomorphisms.
- (c) For integral coefficients, compute the Euler characteristic of $\mathbb{CP}^n$. Explain why the even-dimensional cell attachments make this computation possible without determining the attaching maps on homotopy groups.